C

ACT SUBJECT TEST — MATH 2 • UNIT 2.01

Matrices, Determinants & Vectors

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 C	Q3 B	Q4 A	Q5 A
Q6 B	Q7 A	Q8 A	Q9 D	Q10 C
Q11 D	Q12 B	Q13 C	Q14 D	Q15 D
Q16 A	Q17 B	Q18 A	Q19 A	Q20 A
Q21 D	Q22 B	Q23 C	Q24 D	Q25 D

Matrices, Determinants & Vectors Practice Exercises

1 MCQ

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

- **A** $\langle 4, -3 \rangle$ ✓
- **B** $\langle 1, 1 \rangle$
- **C** $\langle 4, 1 \rangle$
- **D** $\langle 8, -13 \rangle$

WORKED SOLUTION & VERIFICATION

To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$:

$$2\mathbf{u} = 2\langle 3, -4 \rangle = \langle 2(3), 2(-4) \rangle = \langle 6, -8 \rangle$$

Next, add the corresponding components of $2\mathbf{u}$ and $\mathbf{v}$:

$$2\mathbf{u} + \mathbf{v} = \langle 6 + (-2), -8 + 5 \rangle = \langle 4, -3 \rangle$$

- $\langle 1, 1 \rangle$ is incorrect because it represents $\mathbf{u} + \mathbf{v}$, omitting the scalar factor of 2.
- $\langle 4, 1 \rangle$ is incorrect because the scalar 2 was only distributed to the horizontal component.
- $\langle 8, -13 \rangle$ is incorrect because it calculates $2\mathbf{u} - \mathbf{v}$ instead of $2\mathbf{u} + \mathbf{v}$.

2 MCQ

Matrices A and B are defined as $A = \begin{bmatrix} 2 & k \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & k \end{bmatrix}$, where k is a real number. If the determinant of the matrix sum $A + B$ is equal to 17, what is the value of k?

(Note: The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is defined as $ad - bc$.)

A 1

B 2

C 6 ✓

D $\frac{19}{2}$

WORKED SOLUTION & VERIFICATION

Step 1: Compute the matrix sum $A + B$ by adding corresponding entries:

$$A + B = \begin{bmatrix} 2+1 & k+4 \\ -1+2 & 3+k \end{bmatrix} = \begin{bmatrix} 3 & k+4 \\ 1 & k+3 \end{bmatrix}$$

Step 2: Calculate the determinant of $A + B$ using the definition $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$:

$$\det(A + B) = (3)(k+3) - (1)(k+4)$$

Distribute carefully across both terms:

$$\det(A + B) = 3k + 9 - k - 4 = 2k + 5$$

Step 3: Set the determinant equal to 17 and solve for k:

$$2k + 5 = 17 \implies 2k = 12 \implies k = 6$$

Common Misconceptions:

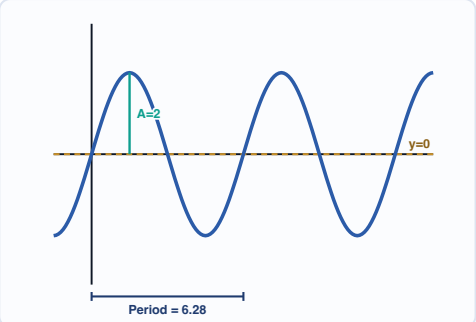
- Choice B (2) results from a sign distribution error when subtracting: $3k + 9 - k + 4 = 2k + 13 = 17 \implies k = 2$.
- Choice A (1) results from incorrectly using $ad + bc$ instead of $ad - bc$: $(3k + 9) + (k + 4) = 4k + 13 = 17 \implies k = 1$.
- Choice D $(\frac{19}{2})$ comes from the false assumption that $\det(A + B) = \det(A) + \det(B)$, which gives $(6 + k) + (k - 8) = 2k - 2 = 17 \implies k = \frac{19}{2}$.

3 MCQ

For all real values of θ, the matrix product

$$[\cos\theta \quad \sin\theta]\begin{bmatrix}6 & 2\\ 2 & 3\end{bmatrix}\begin{bmatrix}\cos\theta\\ \sin\theta\end{bmatrix}$$

evaluates to a 1×1 matrix $[k]$. What is the maximum possible value of k?



A 6

B 7 ✓

C 9

D 14

WORKED SOLUTION & VERIFICATION

Step 1: Perform the matrix multiplication.

First, multiply the 1×2 row vector by the 2×2 matrix:

$$[\cos\theta \quad \sin\theta]\begin{bmatrix}6 & 2\\ 2 & 3\end{bmatrix} = [6\cos\theta + 2\sin\theta \quad 2\cos\theta + 3\sin\theta]$$

Next, multiply this result by the 2×1 column vector:

$$k = (6\cos\theta + 2\sin\theta)\cos\theta + (2\cos\theta + 3\sin\theta)\sin\theta$$

$$k = 6\cos^2\theta + 4\sin\theta\cos\theta + 3\sin^2\theta$$

Step 2: Express the quadratic trigonometric terms using double-angle identities.

Recall the standard identities:

$$\cos^2\theta = \frac{1+\cos(2\theta)}{2}, \quad \sin^2\theta = \frac{1-\cos(2\theta)}{2}, \quad 2\sin\theta\cos\theta = \sin(2\theta)$$

Substitute these into the expression for k:

$$k = 6\left(\frac{1+\cos(2\theta)}{2}\right) + 3\left(\frac{1-\cos(2\theta)}{2}\right) + 2\sin(2\theta)$$

$$k = 3(1+\cos(2\theta)) + \frac{3}{2}(1-\cos(2\theta)) + 2\sin(2\theta)$$

$$k = \frac{9}{2} + \frac{3}{2}\cos(2\theta) + 2\sin(2\theta)$$

Step 3: Maximize the resulting trigonometric expression.

For real constants A and B, the maximum value of $A\cos\phi + B\sin\phi$ is $\sqrt{A^2+B^2}$. Here, $A = \frac{3}{2}$ and $B = 2$:

$$\sqrt{\left(\frac{3}{2}\right)^2 + 2^2} = \sqrt{\frac{9}{4} + \frac{16}{4}} = \sqrt{\frac{25}{4}} = \frac{5}{2}$$

Thus, the maximum value of k is:

$$k_{\max} = \frac{9}{2} + \frac{5}{2} = 7$$

Misconception Analysis:
- Choice A (6) results from ignoring the cross-product term $4 \sin\theta \cos\theta$ and merely selecting the maximum diagonal entry.
- Choice C (9) results from taking the trace (sum of diagonal entries) of the matrix.
- Choice D (14) results from calculating the determinant of the matrix, $6(3) - 2(2) = 14$.

4 MCQ

Given the vectors $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle 1, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

A $\langle 7, 1 \rangle$ ✓

B $\langle 4, 3 \rangle$

C $\langle 8, 6 \rangle$

D $\langle 5, 8 \rangle$

WORKED SOLUTION & VERIFICATION

To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$:

$$2\mathbf{u} = 2\langle 3, -2 \rangle = \langle 2(3), 2(-2) \rangle = \langle 6, -4 \rangle$$

Next, add vector $\mathbf{v}$ by adding corresponding components:

$$2\mathbf{u} + \mathbf{v} = \langle 6, -4 \rangle + \langle 1, 5 \rangle = \langle 6 + 1, -4 + 5 \rangle = \langle 7, 1 \rangle$$

Common Errors:
- $\langle 4, 3 \rangle$ results from computing $\mathbf{u} + \mathbf{v}$ without multiplying $\mathbf{u}$ by 2.
- $\langle 8, 6 \rangle$ results from multiplying both vectors by 2, computing $2(\mathbf{u} + \mathbf{v})$.
- $\langle 5, 8 \rangle$ results from applying the scalar 2 to $\mathbf{v}$ instead of $\mathbf{u}$, computing $\mathbf{u} + 2\mathbf{v}$.

5 MCQ

For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is defined as $ad - bc$. Given the matrices $A = \begin{bmatrix} x & 2 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -4 \\ 3 & x \end{bmatrix}$, the determinant of the matrix $A + B$ is equal to 19. If $x > 0$, what is the value of x?

- **A** 2 ✓
- **B** $\frac{5}{4}$
- **C** 4
- **D** 6

WORKED SOLUTION & VERIFICATION

First, find the matrix sum $A + B$ by adding corresponding entries:

$$A + B = \begin{bmatrix} x+1 & 2+(-4) \\ -1+3 & 3+x \end{bmatrix} = \begin{bmatrix} x+1 & -2 \\ 2 & x+3 \end{bmatrix}$$

Next, calculate the determinant of $A + B$:

$$\det(A+B) = (x+1)(x+3) - (-2)(2)$$

$$\det(A+B) = (x^2 + 4x + 3) - (-4) = x^2 + 4x + 7$$

Set the determinant equal to 19 and solve for x:

$$x^2 + 4x + 7 = 19$$

$$x^2 + 4x - 12 = 0$$

Factoring the quadratic gives:

$$(x+6)(x-2) = 0$$

This yields $x = -6$ or $x = 2$. Since we are given that $x > 0$, the value is $x = 2$.

Common Pitfalls:

- Assuming $\det(A+B) = \det(A) + \det(B)$: $\det(A) = 3x + 2$ and $\det(B) = x + 12$, so setting $4x + 14 = 19$ results in the incorrect answer $x = \frac{5}{4}$.
- Factoring sign error: solving $x^2 + 4x - 12 = 0$ as $(x-6)(x+2) = 0$ yields $x = 6$.

6 MCQ

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the coordinate plane such that $|\mathbf{u}| = 5$, $|\mathbf{v}| = 10$, and the angle θ between them is obtuse ($90^\circ < \theta < 180^\circ$). If the minimum value of $|t\mathbf{u} + \mathbf{v}|$ over all real numbers t is equal to 6, what is the value of the dot product $\mathbf{u} \cdot \mathbf{v}$?

A -50

B -40 ✓

C -30

D 40

WORKED SOLUTION & VERIFICATION

We can solve this problem algebraically or geometrically.

Method 1: Algebraic (Quadratic Optimization)

Consider the square of the magnitude $|t\mathbf{u} + \mathbf{v}|$:

$$|t\mathbf{u} + \mathbf{v}|^2 = (t\mathbf{u} + \mathbf{v}) \cdot (t\mathbf{u} + \mathbf{v}) = t^2|\mathbf{u}|^2 + 2t(\mathbf{u} \cdot \mathbf{v}) + |\mathbf{v}|^2$$

Let $D = \mathbf{u} \cdot \mathbf{v}$. Substituting $|\mathbf{u}| = 5$ and $|\mathbf{v}| = 10$:

$$f(t) = 25t^2 + 2Dt + 100$$

This is a quadratic function in t with leading coefficient $a = 25 > 0$. Its minimum occurs at the vertex:

$$t_{\min} = -\frac{2D}{2(25)} = -\frac{D}{25}$$

Evaluating $f(t)$ at the vertex gives the minimum squared magnitude:

$$f(t_{\min}) = 25\left(-\frac{D}{25}\right)^2 + 2D\left(-\frac{D}{25}\right) + 100 = 100 - \frac{D^2}{25}$$

We are given that the minimum value of $|t\mathbf{u} + \mathbf{v}|$ is 6, so the minimum squared magnitude is $6^2 = 36$:

$$100 - \frac{D^2}{25} = 36 \implies \frac{D^2}{25} = 64 \implies D^2 = 1600$$

Taking the square root gives $D = \pm 40$. Because the angle θ between $\mathbf{u}$ and $\mathbf{v}$ is obtuse ($90^\circ < \theta < 180^\circ$), $\cos\theta < 0$, which implies $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta < 0$. Therefore:

$$\mathbf{u} \cdot \mathbf{v} = -40$$

Method 2: Geometric (Orthogonal Distance)

The set of points with position vectors $t\mathbf{u} + \mathbf{v}$ forms a straight line parallel to $\mathbf{u}$ passing through the endpoint of $\mathbf{v}$. The minimum magnitude $|t\mathbf{u} + \mathbf{v}|$ is the perpendicular distance from the origin to this line.

In the right triangle formed by the origin, the endpoint of $\mathbf{v}$, and the projection point, the hypotenuse is $|\mathbf{v}| = 10$ and the perpendicular leg is 6. Hence, $\sin(180^\circ - \theta) = \sin\theta = \frac{6}{10} = \frac{3}{5}$. Since θ is obtuse, $\cos\theta = -\sqrt{1 - \left(\frac{3}{5}\right)^2} = -\frac{4}{5}$.

Finally, $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta = (5)(10)\left(-\frac{4}{5}\right) = -40$.

Analysis of Distractors:

- -50: Assumes the vectors are directly antiparallel ($\theta = 180^\circ$), yielding $-|\mathbf{u}||\mathbf{v}| = -50$.
- -30: Arises from confusing $\sin\theta$ and $\cos\theta$, erroneously taking $\cos\theta = -\frac{6}{10} = -\frac{3}{5}$, which gives $(50)\left(-\frac{3}{5}\right) = -30$.
- 40: Solves $D^2 = 1600$ correctly but neglects the condition that θ is obtuse, improperly taking the positive root.

7 MCQ

What is the value of the determinant of the matrix $\begin{bmatrix} 7 & -3 \\ 2 & 4 \end{bmatrix}$?

A 34 ✓

B 22

C -29

D -34

WORKED SOLUTION & VERIFICATION

The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is given by the formula:

$$\det = ad - bc$$

Substitute the entries $a = 7$, $b = -3$, $c = 2$, and $d = 4$:

$$\det = (7)(4) - (-3)(2) = 28 - (-6) = 28 + 6 = 34$$

Common misconceptions:
- Calculating $ad + bc$ gives $28 + (-6) = 22$.
- Reversing the subtraction order as $bc - ad$ gives $-6 - 28 = -34$.
- Multiplying along the rows instead of diagonals gives $(7)(-3) - (2)(4) = -21 - 8 = -29$.

8 MCQ

In the standard coordinate plane, vector $\mathbf{u} = \langle 2, -1 \rangle$ and vector $\mathbf{v} = \langle k, 5 \rangle$, where k is a real constant. If the magnitude of the vector $\mathbf{u} + \mathbf{v}$ is equal to 5, which of the following is a possible value of k?

- **A** -5 ✓
- **B** -1
- **C** 3
- **D** 7

WORKED SOLUTION & VERIFICATION

First, find the sum of vectors $\mathbf{u}$ and $\mathbf{v}$ by adding their corresponding components:

$$\mathbf{u} + \mathbf{v} = \langle 2 + k, -1 + 5 \rangle = \langle 2 + k, 4 \rangle$$

Next, use the magnitude formula $\|\langle x, y \rangle\| = \sqrt{x^2 + y^2}$:

$$\sqrt{(2 + k)^2 + 4^2} = 5$$

Square both sides of the equation to eliminate the radical:

$$(2 + k)^2 + 16 = 25$$

$$(2 + k)^2 = 9$$

Take the square root of both sides:

$$2 + k = 3 \quad \text{or} \quad 2 + k = -3$$

$$k = 1 \quad \text{or} \quad k = -5$$

Among the choices, only -5 is listed.

Common traps:

- Solving $(2 + k) + 4 = 5$ by mistakenly adding components linearly instead of using the Pythagorean theorem yields $k = -1$.
- Forgetting the negative square root $2 + k = -3$ causes students to miss $k = -5$.

9 MCQ

In the standard (x, y)-coordinate plane, vectors $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$ have magnitudes $\|\mathbf{u}\| = 13$ and $\|\mathbf{v}\| = 5$. If their dot product is $\mathbf{u} \cdot \mathbf{v} = 25$, and the matrix $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has $\det(M) > 0$, what is the value of $\det(M)$?

A 65

B 25

C 40

D **60** ✓

WORKED SOLUTION & VERIFICATION

We can solve this using either Lagrange's algebraic identity or the geometric interpretation of the dot product and determinant.

Method 1: Lagrange's Identity

For any real numbers a, b, c, d, the identity connecting the determinant and the dot product is:

$$(ad - bc)^2 + (ac + bd)^2 = (a^2 + b^2)(c^2 + d^2)$$

Notice that:

- $\det(M) = ad - bc$
- $\mathbf{u} \cdot \mathbf{v} = ac + bd = 25$
- $\|\mathbf{u}\|^2 = a^2 + b^2 = 13^2 = 169$
- $\|\mathbf{v}\|^2 = c^2 + d^2 = 5^2 = 25$

Substituting these values into the identity:

$$(\det(M))^2 + 25^2 = 169 \times 25$$

$$(\det(M))^2 + 625 = 4{,}225$$

$$(\det(M))^2 = 4{,}225 - 625 = 3{,}600$$

Since $\det(M) > 0$, taking the positive square root gives:

$$\det(M) = \sqrt{3{,}600} = 60$$

Method 2: Geometric Relationship

The dot product satisfies $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$, so:

$$25 = 13 \times 5 \times \cos\theta = 65\cos\theta \implies \cos\theta = \frac{25}{65} = \frac{5}{13}$$

Using $\sin^2\theta + \cos^2\theta = 1$:

$$\sin\theta = \sqrt{1 - \left(\frac{5}{13}\right)^2} = \sqrt{\frac{144}{169}} = \frac{12}{13}$$

The magnitude of $\det(M)$ is equal to the area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$:

$$|\det(M)| = \|\mathbf{u}\|\|\mathbf{v}\| \sin\theta = 13 \times 5 \times \frac{12}{13} = 60$$

Since $\det(M) > 0$, $\det(M) = 60$.

Misconception Analysis:

- Choice A (25) confuses the determinant $ad - bc$ with the dot product $ac + bd$.
- Choice B (40) erroneously computes $\|\mathbf{u}\|\|\mathbf{v}\| - (\mathbf{u} \cdot \mathbf{v}) = 65 - 25 = 40$.

- Choice D (65) computes the product of magnitudes $\|\mathbf{u}\|\|\mathbf{v}\| = 13 \times 5 = 65$, which only equals the determinant if the vectors are orthogonal.

10 MCQ

If matrix $A = \begin{bmatrix} 3 & -2 \\ 0 & 4 \end{bmatrix}$ and matrix $B = \begin{bmatrix} -1 & 5 \\ 2 & -3 \end{bmatrix}$, what is the matrix $2A + B$?

A $\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$

B $\begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix}$

C $\begin{bmatrix} 5 & 1 \\ 2 & 5 \end{bmatrix}$ ✓

D $\begin{bmatrix} 7 & -9 \\ -2 & 11 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

To find the matrix $2A + B$, first perform scalar multiplication by multiplying each entry of matrix A by 2:

$$2A = 2\begin{bmatrix} 3 & -2 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 2(3) & 2(-2) \\ 2(0) & 2(4) \end{bmatrix} = \begin{bmatrix} 6 & -4 \\ 0 & 8 \end{bmatrix}$$

Next, add the corresponding entries of $2A$ and B:

$$2A + B = \begin{bmatrix} 6 & -4 \\ 0 & 8 \end{bmatrix} + \begin{bmatrix} -1 & 5 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 6 + (-1) & -4 + 5 \\ 0 + 2 & 8 + (-3) \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 2 & 5 \end{bmatrix}$$

Common Errors:

- $\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$ is obtained by simply computing $A + B$ without multiplying by the scalar 2.
- $\begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix}$ results from mistakenly multiplying only the first entry of A by 2.
- $\begin{bmatrix} 7 & -9 \\ -2 & 11 \end{bmatrix}$ results from computing $2A - B$ instead of $2A + B$.

11 MCQ

If $A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 0 & 4 \end{bmatrix}$, find AB.

A $\begin{bmatrix} -2 & 1 \\ 0 & -4 \end{bmatrix}$

B $\begin{bmatrix} -2 & 4 \\ 0 & -4 \end{bmatrix}$

C $\begin{bmatrix} -1 & -1 \\ 0 & 3 \end{bmatrix}$

D $\begin{bmatrix} -2 & 2 \\ 0 & -4 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

$(AB)_{11} = -2 \cdot 1 + 0 \cdot 0 = -2$, and similarly for the other three entries: $AB = \begin{bmatrix} -2 & 2 \\ 0 & -4 \end{bmatrix}$.

12 MCQ

If $A = \begin{bmatrix} 1 & -2 \\ 0 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 2 & 4 \end{bmatrix}$, find AB.

A $\begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$

B $\begin{bmatrix} -1 & -4 \\ -4 & -8 \end{bmatrix}$ ✓

C $\begin{bmatrix} 3 & -14 \\ 2 & -12 \end{bmatrix}$

D $\begin{bmatrix} -1 & -2 \\ -4 & -8 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

$(AB)_{11} = 1 \cdot 3 + -2 \cdot 2 = -1$, and similarly for the other three entries: $AB = \begin{bmatrix} -1 & -4 \\ -4 & -8 \end{bmatrix}$.

13 MCQ

If $A = \begin{bmatrix} -4 & -3 \\ -3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -4 \\ 1 & 4 \end{bmatrix}$, find AB.

A $\begin{bmatrix} 0 & -7 \\ -2 & 2 \end{bmatrix}$

B $\begin{bmatrix} -4 & -4 \\ -16 & -11 \end{bmatrix}$

C $\begin{bmatrix} -19 & 4 \\ -14 & 4 \end{bmatrix}$ ✓

D $\begin{bmatrix} -19 & 6 \\ -14 & 4 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

$(AB)_{11} = -4 \cdot 4 + -3 \cdot 1 = -19$, and similarly for the other three entries: $AB = \begin{bmatrix} -19 & 4 \\ -14 & 4 \end{bmatrix}$.

14 MCQ

If $A = \begin{bmatrix} -3 & 1 \\ -3 & -5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 6 \\ 11 \end{bmatrix}$

B $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$

C $\begin{bmatrix} 6 \\ 5 \end{bmatrix}$

D $\begin{bmatrix} 5 \\ 11 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-3(-2) + 1(-1) = 5$, second entry $-3(-2) + -5(-1) = 11$, so $A\mathbf{v} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$.

15 MCQ

If $A = \begin{bmatrix} -5 & 0 \\ -2 & 5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 15 \\ 25 \end{bmatrix}$

B $\begin{bmatrix} 15 \\ 20 \end{bmatrix}$

C $\begin{bmatrix} 16 \\ 26 \end{bmatrix}$

D $\begin{bmatrix} 15 \\ 26 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-5(-3) + 0(4) = 15$, second entry $-2(-3) + 5(4) = 26$, so $A\mathbf{v} = \begin{bmatrix} 15 \\ 26 \end{bmatrix}$.

16 MCQ

If $A = \begin{bmatrix} -4 & 0 \\ 3 & 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} -12 \\ 14 \end{bmatrix}$ ✓

B $\begin{bmatrix} -11 \\ 14 \end{bmatrix}$

C $\begin{bmatrix} -12 \\ 13 \end{bmatrix}$

D $\begin{bmatrix} -12 \\ 5 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-4(3) + 0(5) = -12$, second entry $3(3) + 1(5) = 14$, so $A\mathbf{v} = \begin{bmatrix} -12 \\ 14 \end{bmatrix}$.

17 MCQ

If $A = \begin{bmatrix} -3 & -2 \\ -3 & -3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ -6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 0 \\ 18 \end{bmatrix}$

B $\begin{bmatrix} 12 \\ 18 \end{bmatrix}$ ✓

C $\begin{bmatrix} 13 \\ 18 \end{bmatrix}$

D $\begin{bmatrix} 12 \\ 17 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-3(0) + -2(-6) = 12$, second entry $-3(0) + -3(-6) = 18$, so $A\mathbf{v} = \begin{bmatrix} 12 \\ 18 \end{bmatrix}$.

18 MCQ

Suppose that $A = \begin{bmatrix} -3 & 0 \\ 1 & 5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, find $A\mathbf{v}$.

- **A** $\begin{bmatrix} -15 \\ 35 \end{bmatrix}$ ✓
- **B** $\begin{bmatrix} -14 \\ 35 \end{bmatrix}$
- **C** $\begin{bmatrix} -15 \\ 34 \end{bmatrix}$
- **D** $\begin{bmatrix} -15 \\ 30 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-3(5) + 0(6) = -15$, second entry $1(5) + 5(6) = 35$, so $A\mathbf{v} = \begin{bmatrix} -15 \\ 35 \end{bmatrix}$.

19 MCQ

Suppose that $A = \begin{bmatrix} -2 & -1 \\ -3 & -4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ -6 \end{bmatrix}$, find $A\mathbf{v}$.

- **A** $\begin{bmatrix} 6 \\ 24 \end{bmatrix}$ ✓
- **B** $\begin{bmatrix} 7 \\ 24 \end{bmatrix}$
- **C** $\begin{bmatrix} 6 \\ 23 \end{bmatrix}$
- **D** $\begin{bmatrix} 0 \\ 24 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-2(0) + -1(-6) = 6$, second entry $-3(0) + -4(-6) = 24$, so $A\mathbf{v} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$.

20 MCQ

If $A = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$, find $A\mathbf{v}$.

- **A** $\begin{bmatrix} 2 \\ -2 \end{bmatrix}$ ✓
- **B** $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$
- **C** $\begin{bmatrix} -2 \\ -4 \end{bmatrix}$
- **D** $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $2(-1) + -2(-2) = 2$, second entry $-2(-1) + 2(-2) = -2$, so $A\mathbf{v} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$.

21 MCQ

If $A = \begin{bmatrix} -2 & -2 \\ -5 & 5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$

B $\begin{bmatrix} 7 \\ 15 \end{bmatrix}$

C $\begin{bmatrix} 6 \\ 14 \end{bmatrix}$

D $\begin{bmatrix} 6 \\ 15 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-2(-3) + -2(0) = 6$, second entry $-5(-3) + 5(0) = 15$, so $A\mathbf{v} = \begin{bmatrix} 6 \\ 15 \end{bmatrix}$.

22 MCQ

If $A = \begin{bmatrix} 0 & -1 \\ -4 & -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 0 \\ -4 \end{bmatrix}$

B $\begin{bmatrix} -4 \\ 0 \end{bmatrix}$ ✓

C $\begin{bmatrix} -3 \\ 0 \end{bmatrix}$

D $\begin{bmatrix} -4 \\ -1 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $0(-1) + -1(4) = -4$, second entry $-4(-1) + -1(4) = 0$, so $A\mathbf{v} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}$.

23 MCQ

If $A = \begin{bmatrix} 3 & 1 \\ 5 & 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ -6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} -8 \\ -23 \end{bmatrix}$

B $\begin{bmatrix} -9 \\ -24 \end{bmatrix}$

C $\begin{bmatrix} -9 \\ -23 \end{bmatrix}$ ✓

D $\begin{bmatrix} -3 \\ -18 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $3(-1) + 1(-6) = -9$, second entry $5(-1) + 3(-6) = -23$, so $A\mathbf{v} = \begin{bmatrix} -9 \\ -23 \end{bmatrix}$.

24 MCQ

If $A = \begin{bmatrix} -4 & -1 \\ -3 & 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -2 \\ -6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 14 \\ -19 \end{bmatrix}$

B $\begin{bmatrix} 8 \\ -24 \end{bmatrix}$

C $\begin{bmatrix} 15 \\ -18 \end{bmatrix}$

D $\begin{bmatrix} 14 \\ -18 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-4(-2) + -1(-6) = 14$, second entry $-3(-2) + 4(-6) = -18$, so $A\mathbf{v} = \begin{bmatrix} 14 \\ -18 \end{bmatrix}$.

25 MCQ

Assuming that $A = \begin{bmatrix} -4 & 4 \\ 1 & 0 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 5 \\ 5 \end{bmatrix}$

B $\begin{bmatrix} 4 \\ 4 \end{bmatrix}$

C $\begin{bmatrix} -20 \\ 0 \end{bmatrix}$

D $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

Row-by-vector: first entry $-4(5) + 4(6) = 4$, second entry $1(5) + 0(6) = 5$, so $A\mathbf{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$.